

Bash Scripting

```
[root@ip-172-31-0-121 ~]# vi test
[root@ip-172-31-0-121 ~]# ll
total 8
-rw-r--r--. 1 root root 40 Mar 28 11:40 alphabet
-rw-r--r--. 1 root root 32 Apr  2 09:36 test
[root@ip-172-31-0-121 ~]# vi test
[root@ip-172-31-0-121 ~]# chmod 777 test
[root@ip-172-31-0-121 ~]# ./test
Mem:           total        used          free      shared  buff/cache   available
Swap:              0              0              0
UID            PID     PPID  C  STIME TTY          TIME CMD
root             1         0  0  09:31 ?        00:00:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root             2         0  0  09:31 ?        00:00:00 [kthreadd]
root             3         2  0  09:31 ?        00:00:00 [pool_workqueue_release]
```

```
root@ip-172-31-0-121:~
#!/bin/bash

free -m
ps -ef
df
|
~
```

Types of shells

```
[root@ip-172-31-0-121 ~]# echo $0
bash
[root@ip-172-31-0-121 ~]# cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/csh
/bin/tcsh
/usr/bin/csh
/usr/bin/tcsh
[root@ip-172-31-0-121 ~]#
```

```
[root@ip-172-31-0-121 ~]# ech $0
bash: ech: command not found
[root@ip-172-31-0-121 ~]# echo $0
bash
[root@ip-172-31-0-121 ~]# cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/csh
/bin/tcsh
/usr/bin/csh
/usr/bin/tcsh
[root@ip-172-31-0-121 ~]# sh
sh-5.2# echo $0
sh
sh-5.2# exit
exit
[root@ip-172-31-0-121 ~]#
```

```
[root@ip-172-31-0-121 ~]# csh
[root@ip-172-31-0-121 ~]# echo $0
csh
[root@ip-172-31-0-121 ~]# tcsh
[root@ip-172-31-0-121 ~]# echo $0
tcsh
[root@ip-172-31-0-121 ~]# bash
[root@ip-172-31-0-121 ~]# echo $0
bash
```

Creating mkdir -p and save with script to save the data at once place

```
[root@ip-172-31-0-121 ~]# pwd
/root
[root@ip-172-31-0-121 ~]# mkdir -p /arun/scripts
[root@ip-172-31-0-121 ~]# cd /arun/scripts
[root@ip-172-31-0-121 scripts]# ls
[root@ip-172-31-0-121 scripts]# pwd
/arun/scripts
[root@ip-172-31-0-121 scripts]# vi hello world.bash
2 files to edit
[root@ip-172-31-0-121 scripts]# vi hello world.bash
2 files to edit
[root@ip-172-31-0-121 scripts]# ll
total 4
-rw-r--r--. 1 root root 30 Apr  2 09:53 hello
[root@ip-172-31-0-121 scripts]# chmod 777 hello world
chmod: cannot access 'world': No such file or directory
[root@ip-172-31-0-121 scripts]# chmod 777 hello
[root@ip-172-31-0-121 scripts]# ./hello
hello world
[root@ip-172-31-0-121 scripts]#
```

1. Basic Shell Scripts:

Output to screen

#!/bin/bash

Simple output script

echo "Hello World"

```
[root@ip-172-31-0-121 scripts]# pwd
/arun/scripts
[root@ip-172-31-0-121 scripts]# ll
total 4
-rwxrwxrwx. 1 root root 30 Apr  2 09:53 hello
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash
echo hello world
```

```
~
~
~
~
~
~
~
```

```
#!/bin/bash
#Purpose: Testing script format
#Date: 02/04/2026
#Modification: 02/04/2026
```

```
a ='Hello World'
```

```
echo $a
```

```
[root@ip-172-31-0-121 scripts]# cat helloworld.bash
#!/bin/bash
#Purpose: Testing script format
#Date: 02/04/2026
#Modification: 02/04/2026

a ='Hello World'

echo $a

[root@ip-172-31-0-121 scripts]#
```

- **Commands .bash**

```
-rw-r--r--. 1 root root  22 Apr  2 10:14 abc.bash
-rw-r--r--. 1 root root  47 Apr  2 10:26 commands.bash
-rwxr-xr-x. 1 root root 117 Apr  2 10:20 helloworld.bash
[root@ip-172-31-0-121 scripts]# ./commands.bash
bash: ./commands.bash: Permission denied
[root@ip-172-31-0-121 scripts]# chmod 755 commands.bash
[root@ip-172-31-0-121 scripts]# ./commands.bash
/arun/scripts
abc.bash  commands.bash  helloworld.bash
root
Thu Apr  2 10:26:57 UTC 2026
    April 2026
Su Mo Tu We Th Fr Sa
      1  2  3  4
  5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
```

```

root@ip-172-31-0-121 scripts]# pwd
/arun/scripts
root@ip-172-31-0-121 scripts]# ls
a abc.bash b c commands.bash helloworld.bash
root@ip-172-31-0-121 scripts]# whoami
root
root@ip-172-31-0-121 scripts]#

```

- **admin.bash**

```

#!/bin/bash
top
df -h
free -m
uptime
iostat

```

```

[root@ip-172-31-0-121 scripts]# vi admin.bash
[root@ip-172-31-0-121 scripts]# chmod 755 admin.bash
[root@ip-172-31-0-121 scripts]# ./admin.bash
top - 10:34:37 up 1:03, 2 users, load average: 0.12, 0.03, 0.01
Tasks: 124 total, 1 running, 123 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.3 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 914.1 total, 67.2 free, 605.4 used, 241.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 171.2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 1729 jenkins    20   0 3095016 365208 25340 S   0.7   39.0   0:39.48 java
 1736 root        20   0 1807632 37644 23256 S   0.3   4.0   0:02.57 containerd
 1771 ec2-user    20   0 3000724 92636 22924 S   0.3   9.9   0:10.44 java
 1857 apache     20   0 1544808 7722 4412 S   0.2   0.8   0:01.21 httpd

```

- **variables.bash**

```

[root@ip-172-31-0-121 scripts]# vi variables.bash
[root@ip-172-31-0-121 scripts]# chmod 755 variables.bash
[root@ip-172-31-0-121 scripts]# ./variables.bash
Run Variables tasks

/arun/scripts
a abc.bash admin.bash b c commands.bash helloworld.bash variables.bash
root
Thu Apr  2 10:37:12 UTC 2026

                                2026

      January                February                March
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
      1  2  3          1  2  3  4  5  6  7      1  2  3  4  5  6  7
  4  5  6  7  8  9 10      8  9 10 11 12 13 14      8  9 10 11 12 13 14
11 12 13 14 15 16 17      15 16 17 18 19 20 21      15 16 17 18 19 20 21
18 19 20 21 22 23 24      22 23 24 25 26 27 28      22 23 24 25 26 27 28
25 26 27 28 29 30 31                                29 30 31

      April                  May                  June

```

- **input.bash**

```
[root@ip-172-31-0-121 scripts]#  
[root@ip-172-31-0-121 scripts]# vi input.bash  
[root@ip-172-31-0-121 scripts]# chmod 755 input.bash  
[root@ip-172-31-0-121 scripts]# ./input.bash  
Hello, my name is arun  
  
what is your name?  
Sai  
  
Hello Sai  
  
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash  
echo Hello, my name is arun  
echo  
echo what is your name?  
read name  
echo  
echo Hello $name  
echo
```

- **output. Bash**

```
[root@ip-172-31-0-121 scripts]# ./output.bash  
Hello, my name is ip-172-31-0-121.ec2.internal  
  
what is your name?  
kumar  
  
Hello kumar  
  
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash  
  
#we need to use the ticks if you want to use this in echo or else not required  
a=`hostname`  
echo Hello, my name is $a  
echo  
echo what is your name?  
read b  
echo  
echo Hello $b  
echo
```

2. Defining Tasks

```
#!/bin/bash
# Define small tasks

whoami
echo
pwd
echo
hostname
echo
ls -ltr
echo
```

```
[root@ip-172-31-0-121 scripts]# chmod 445 define.bash
[root@ip-172-31-0-121 scripts]# ./define.bash
root

/arun/scripts

ip-172-31-0-121.ec2.internal

total 32
-rw-r--r--. 1 root root  22 Apr  2 10:14 abc.bash
-rwxr-xr-x. 1 root root 117 Apr  2 10:20 helloworld.bash
-rwxr-xr-x. 1 root root  47 Apr  2 10:26 commands.bash
-rw-r--r--. 1 root root   0 Apr  2 10:29 c
-rw-r--r--. 1 root root   0 Apr  2 10:29 b
-rw-r--r--. 1 root root   0 Apr  2 10:29 a
-rwxr-xr-x. 1 root root  45 Apr  2 10:35 admin.bash
-rwxr-xr-x. 1 root root 119 Apr  2 10:36 variables.bash
-rwxr-xr-x. 1 root root 107 Apr  2 10:38 input.bash
-rwxr-xr-x. 1 root root 198 Apr  2 10:40 output.bash
-r--r--r-x. 1 root root  84 Apr  2 10:44 define.bash

[root@ip-172-31-0-121 scripts]#
```

3. Defining variables

```
#!/bin/bash
# Example of defining variables

a=Arun
b=Katti
c='Linux class'

echo "My first name is $a"
echo "My surname is $b"
echo 'My surname is $c'
```

```
[root@ip-172-31-0-121 scripts]# vi definevariabeles.bash
[root@ip-172-31-0-121 scripts]# chmod 755 definevariables.bash
chmod: cannot access 'definevariables.bash': No such file or directory
[root@ip-172-31-0-121 scripts]# chmod 755 definevariabeles.bash
[root@ip-172-31-0-121 scripts]# ./definevariabeles.bash
./definevariabeles.bash: line 6: class': command not found
My first name is Arun
My surname is Katti
'My surname is '
[root@ip-172-31-0-121 scripts]#
```

4. Scripts to run commands within

```
#!/bin/bash
# Script to run commands within

clear
echo "Hello `whoami`"
echo
echo "Today is `date`"
echo
echo "Number of user login: `who | wc -l`"
echo
```



```
Hello root
```

```
Today is Thu Apr  2 10:49:46 UTC 2026
```

```
Number of user login: 2
```

```
[root@ip-172-31-0-121 scripts]#
```

5. Read input and perform a task

```
#!/bin/bash
# This script will rename a file

echo Enter the file name to be renamed
read oldfilename
echo Enter the new file name
read newfilename
mv $oldfilename $newfilename
echo The file has been renamed as $newfilename
```

```
[root@ip-172-31-0-121 scripts]# mkdir -p devops
[root@ip-172-31-0-121 scripts]# ls
1 a          backup.sh    define.bash   forloop.sh    read          specify.sh
2 abc.bash   c           definevariabeles.bash helloworld.bash read.sh       users.sh
3 admin.bash case.sh      devops        if.sh         response.sh   value.bash
4 arun       commands.bash dowhile.sh    input.bash    seconds.sh   variables.bash
5 b          create.sh    error.sh      output.bash    simple.sh    within.sh
[root@ip-172-31-0-121 scripts]# ./read.sh
Enter the file name to be renamed
devops
Enter the new file name
linux
The file has been renamed as linux
[root@ip-172-31-0-121 scripts]# ls
1 a          backup.sh    define.bash   helloworld.bash read          specify.sh
2 abc.bash   c           definevariabeles.bash if.sh         read.sh       users.sh
3 admin.bash case.sh      dowhile.sh    input.bash    response.sh   value.bash
4 arun       commands.bash error.sh      linux         seconds.sh    variables.bash
5 b          create.sh    forloop.sh    output.bash    simple.sh    within.sh
```

case Scripts:

```
echo
echo Please chose one of the options below
echo
echo 'a = Display Date and Time'
echo 'b = List file and directories'
echo 'c = List users logged in'
echo 'd = Check System uptime'
echo

    read choices

    case $choices in

a) date;;
b) ls;;
c) who;;
d) uptime;;
*) echo Invalid choice - Bye.

    esac
```

```
[root@ip-172-31-0-121 scripts]# vi case.sh
[root@ip-172-31-0-121 scripts]# chmod 755 case.sh
[root@ip-172-31-0-121 scripts]# ./case.sh

Please chose one of the options below

a = Display Date and Time
b = List file and directories
c = List users logged in
d = Check System uptime

b
a      admin.bash  c      commands.bash  definevariabeles.bash  input.bash  read  variables.bash
abc.bash  b      case.sh  define.bash  helloworld.bash  output.bash  read.sh  within.sh
[root@ip-172-31-0-121 scripts]#
```

- This script will look at your current day and tell you the state of the backup

```
#!/bin/bash

NOW=$(date +%a)
case $NOW in
    Mon)
        echo "Full backup";;
    Tue|Wed|Thu|Fri)
        echo "Partial backup";;
    Sat|Sun)
        echo "No backup";;
    *) ;;
esac

~
```

```
[root@ip-172-31-0-121 scripts]# vi backup.sh
[root@ip-172-31-0-121 scripts]# chmod 765 backup.sh
[root@ip-172-31-0-121 scripts]# ./backup.sh
Partial backup
[root@ip-172-31-0-121 scripts]#
```

- **do-while Script**

Script to run for a number of times

```
#!/bin/bash

c=1
while [ $c -le 5 ]
do
    echo "Welcone $c times"
    (( c++ ))
done
```

```

[root@ip-172-31-0-121 scripts]# vi dowhile.sh
[root@ip-172-31-0-121 scripts]# chmod 755 dowhile.sh
[root@ip-172-31-0-121 scripts]# ./dowhile.sh
Welcone 1 times
Welcone 2 times
Welcone 3 times
Welcone 4 times
Welcone 5 times
[root@ip-172-31-0-121 scripts]# vi dowhile.sh
[root@ip-172-31-0-121 scripts]# vi dowhile.sh
[root@ip-172-31-0-121 scripts]# chmod 755 dowhile.sh
[root@ip-172-31-0-121 scripts]# ./dowhile.sh
Welcone 1 times
Welcone 2 times
Welcone 3 times
[root@ip-172-31-0-121 scripts]#

```

- Script to run for a number of seconds

```

#!/bin/bash

count=5
num=5
while [ $count -lt 5 ]
do
    echo
    echo $num seconds left to stop this process $1
    echo
    sleep 1

    num=`expr $num - 1`
    count=`expr $count + 1`
done
echo
echo $1 process is stopped!!!
echo

```

```

[root@ip-172-31-0-121 scripts]# vi seconds.sh
[root@ip-172-31-0-121 scripts]# chmod 755 seconds.sh
[root@ip-172-31-0-121 scripts]# ./seconds.sh

10 seconds left to stop this process

9 seconds left to stop this process

8 seconds left to stop this process

7 seconds left to stop this process

6 seconds left to stop this process

process is stopped!!!

```

- **for loop Scripts:**

Simple for loop output

```
#!/bin/bash

for i in 1 2 3 4 5
do
echo "Welcome $i times"
done
```

```
[root@ip-172-31-0-121 scripts]# vi forloop.sh
[root@ip-172-31-0-121 scripts]# chmod 745 forlop.sh
chmod: cannot access 'forlop.sh': No such file or directory
[root@ip-172-31-0-121 scripts]# chmod 745 forloop.sh
[root@ip-172-31-0-121 scripts]# ./forloop.sh
Welcome 1 times
Welcome 2 times
Welcome 3 times
Welcome 4 times
Welcome 5 times
[root@ip-172-31-0-121 scripts]#
```

- Simple for loop output

```
#!/bin/bash

for i in eat run jump play
do
echo See aman $i
done
```

```
[root@ip-172-31-0-121 scripts]# vi simple.sh
[root@ip-172-31-0-121 scripts]# chmod 654 simple.sh
[root@ip-172-31-0-121 scripts]# vi simple.sh
[root@ip-172-31-0-121 scripts]# ./simple.sh
See aman eat
See aman run
See aman jump
See aman play
[root@ip-172-31-0-121 scripts]#
```

- for loop to create 5 files named 1-5

```
#!/bin/bash

for i in {1..5}
do
    touch $i
done
```

```
[root@ip-172-31-0-121 scripts]# vi create.sh
[root@ip-172-31-0-121 scripts]# chmod 755 create.sh
[root@ip-172-31-0-121 scripts]# ./create.sh
[root@ip-172-31-0-121 scripts]# vi create.sh
[root@ip-172-31-0-121 scripts]# ls
1 4 abc.bash backup.sh commands.bash definevariable.bash helloworld.bash read simple.sh
2 5 admin.bash c create.sh dowhile.sh input.bash read.sh variables.bash
3 a b case.sh define.bash forloop.sh output.bash seconds.sh within.sh
[root@ip-172-31-0-121 scripts]# create.sh
bash: create.sh: command not found
[root@ip-172-31-0-121 scripts]# ./create.sh
[root@ip-172-31-0-121 scripts]# vi create.sh
[root@ip-172-31-0-121 scripts]# chmod 755 create.sh
[root@ip-172-31-0-121 scripts]# ./create.sh
[root@ip-172-31-0-121 scripts]# ls
1 5 arun case.sh definevariable.bash input.bash seconds.sh
2 a b commands.bash dowhile.sh output.bash simple.sh
3 abc.bash backup.sh create.sh forloop.sh read variables.bash
4 admin.bash c define.bash helloworld.bash read.sh within.sh
[root@ip-172-31-0-121 scripts]#
```

- Specify days in for loop

```
[root@ip-172-31-0-121 scripts]# vi specify.sh
[root@ip-172-31-0-121 scripts]# chmod 755 specify.sh
[root@ip-172-31-0-121 scripts]# ./specify.sh
Weekday 1 : Mon
Weekday 2 : Tue
Weekday 3 : Wed
Weekday 4 : Thu
Weekday 5 : Fri
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash

i=1
for day in Mon Tue Wed Thu Fri
do
    echo "Weekday $((i++)) : $day"
done
```

- List all users one by one from /etc/passwd file

```
#!/bin/bash

i=1
for username in `awk -F: '{print $1}' /etc/passwd`
do
    echo "Username $((i++)) : $username"
done
```

```
[root@ip-172-31-0-121 scripts]# vi users.sh
[root@ip-172-31-0-121 scripts]# chmod 745 users.sh
[root@ip-172-31-0-121 scripts]# ./users.sh
Username 1 : root
Username 2 : bin
Username 3 : daemon
Username 4 : adm
Username 5 : lp
Username 6 : sync
Username 7 : shutdown
Username 8 : halt
Username 9 : mail
Username 10 : operator
Username 11 : games
Username 12 : ftp
Username 13 : nobody
Username 14 : dbus
Username 15 : systemd-network
Username 16 : systemd-oom
Username 17 : systemd-resolve
Username 18 : rpc
Username 19 : libstoragegmt
```

- **If-then Scripts:**

Check the variable

```
#!/bin/bash

count=100
if [ $count -eq 10 ]
then
    echo Count is 10
else
    echo Count is not 10
fi
```

```
[root@ip-172-31-0-121 scripts]# vi if.sh
[root@ip-172-31-0-121 scripts]# chmod 445 if.sh
[root@ip-172-31-0-121 scripts]# ./if.sh
Count is 100
[root@ip-172-31-0-121 scripts]# vi if.sh
[root@ip-172-31-0-121 scripts]# chmod 445 if.sh
[root@ip-172-31-0-121 scripts]# ./if.sh
Count is 10
[root@ip-172-31-0-121 scripts]#
```

- Check if a file error.txt exist

```
#!/bin/bash

clear
if [ -e /home/arun/error.txt ]

    then
        echo "File exist"
    else
        echo "File does not exist"
fi
```



```
File does not exist
[root@ip-172-31-0-121 scripts]#
```

- Check if a variable value is met

```
#!/bin/bash

a=`date | awk '{print $1}'`

if [ "$a" == Thu ]

    then
    echo Today is $a
    else
    echo Today is not Thursday
fi
```

```
[root@ip-172-31-0-121 scripts]# vi value.bash
[root@ip-172-31-0-121 scripts]# chmod 445 value.sh
chmod: cannot access 'value.sh': No such file or directory
[root@ip-172-31-0-121 scripts]# chmod 445 value.bash
[root@ip-172-31-0-121 scripts]# ./value.bash
Today is Thu
[root@ip-172-31-0-121 scripts]#
```

- Check the response and then output

```
[root@ip-172-31-0-121 scripts]# vi value.bash
[root@ip-172-31-0-121 scripts]# chmod 445 value.sh
chmod: cannot access 'value.sh': No such file or directory
[root@ip-172-31-0-121 scripts]# chmod 445 value.bash
[root@ip-172-31-0-121 scripts]# ./value.bash
Today is Thu
[root@ip-172-31-0-121 scripts]#
```

What is your name?

Arun

Hello Arun sir

Do you like working in IT? (y/n)

y

You are cool

[root@ip-172-31-0-121 scripts]#